

Derivatives Pricing & Hedging Engine

Detailed Quantitative Finance Study Material

Scope: theory, derivations, examples, assumptions, exceptions, numerical methods, implementation guidance, validation and interview preparation.

Target: a project that demonstrates quantitative finance understanding—not merely formula implementation.

Important: For every major formula, learn four things: derivation, assumptions, economic interpretation, and failure/extension cases.

1. Mathematical Foundations

Probability

Random variables, expectation, variance, covariance, conditional expectation and independence are the basic language of pricing. Expectation is linear even without independence: $E[aX+bY]=aE[X]+bE[Y]$.

$$E[aX+bY]=aE[X]+bE[Y]$$

Variance

$\text{Var}(X)=E[X^2]-E[X]^2$. $\text{Cov}(X,Y)=E[XY]-E[X]E[Y]$. Correlation is covariance divided by $\sigma_X\sigma_Y$.

$$\text{Var}(X)=E[X^2]-E[X]^2; \text{Cov}(X,Y)=E[XY]-E[X]E[Y]$$

Normal distribution

$Z \sim N(0,1)$. The density is $\phi(z)=\exp(-z^2/2)/\sqrt{2\pi}$, and $N(z)$ is its CDF. Black-Scholes is built around normal/lognormal transformations.

Lognormal distribution

If $\ln(S_T)$ is normal, S_T is lognormal and positive. GBM generates a lognormal terminal price. This is convenient but cannot represent jumps or negative prices.

LLN and CLT

The Law of Large Numbers explains Monte Carlo convergence of sample means. The CLT explains approximate normality of the estimator and motivates confidence intervals.

Calculus

Know partial derivatives, chain rule, Taylor expansion and integration. Greeks are derivatives; Black-Scholes uses a second-order stochastic Taylor expansion.

Taylor expansion

$f(x+h)=f(x)+hf'(x)+\frac{1}{2}h^2f''(x)+\dots$. Delta is first-order sensitivity; Gamma is curvature.

$$f(x+h) \approx f(x) + hf'(x) + \frac{1}{2}h^2f''(x)$$

Interview

Why does Monte Carlo error scale as $1/\sqrt{N}$? Because the standard error of a sample mean decreases as the inverse square root of the sample size.

2. No-Arbitrage, Forwards & Options

No-arbitrage

An arbitrage strategy has no net initial cost and cannot lose, with a positive gain in at least one state. In idealized pricing, identical future cash flows must have identical prices.

Forward price

With continuous dividend yield q : $F_0 = S_0 \exp((r-q)T)$. Financing costs increase the forward price; dividends reduce it.

$$F_0 = S_0 \exp((r-q)T)$$

Cash-and-carry

Replicate a long forward by buying the asset and financing it. Hold through maturity and account for dividends. This gives the no-arbitrage forward.

Call and put

Call payoff = $\max(S_T - K, 0)$. Put payoff = $\max(K - S_T, 0)$. European exercise is only at maturity; American exercise is allowed before maturity.

Moneyness

ITM/ATM/OTM describes strike relative to spot. It matters for Delta, Gamma, Vega, liquidity and implied volatility.

Intrinsic/time value

Intrinsic value is immediate exercise value. Time value reflects remaining optionality and uncertainty.

Exception

American calls on a non-dividend-paying stock generally should not be exercised early under standard assumptions. Dividends, rates and contract details can change the result.

Interview

Why is an option valuable before expiry? Because convex payoff plus uncertainty creates value in retaining the right to exercise.

3. Put-Call Parity & Bounds

Parity

For European options: $C - P = S_0 \exp(-qT) - K \exp(-rT)$.

$$C - P = S_0 \exp(-qT) - K \exp(-rT)$$

Derivation

Long call + discounted strike and long put + stock have identical terminal payoffs. Therefore their prices must match by no-arbitrage.

Bounds

Call: $\max(0, S_0 e^{-qT} - K e^{-rT}) \leq C \leq S_0 e^{-qT}$. Put: $\max(0, K e^{-rT} - S_0 e^{-qT}) \leq P \leq K e^{-rT}$.

Example

$S=100, K=100, r=5\%, q=0, T=1$ gives $C - P \approx 100 - 100e^{-0.05} \approx 4.877$. If $C=10.45$, parity implies $P \approx 5.573$.

Implementation

Use parity as an automated invariant. Also test bounds and limiting cases.

Exception

Bid-ask spreads, discrete dividends, funding, transaction costs, settlement conventions and early exercise mean observed prices may not satisfy the ideal identity exactly.

Interview

Can parity be used for arbitrage? Yes, if violations exceed realistic transaction/funding costs and the contracts are truly comparable.

4. Binomial Model

One-period replication

Let $S_u = S_0u$ and $S_d = S_0d$, with option values V_u, V_d . Solve $\Delta S_u + B e^{r\Delta t} = V_u$ and $\Delta S_d + B e^{r\Delta t} = V_d$.

Hedge ratio

$\Delta = (V_u - V_d) / (S_u - S_d)$. The hedge ratio emerges from replication.

Risk-neutral probability

$p^* = [e^{(r-q)\Delta t} - d] / (u - d)$. Then $V_0 = e^{-r\Delta t} [p^* V_u + (1 - p^*) V_d]$.

CRR tree

$u = e^{\sigma\sqrt{\Delta t}}$, $d = e^{-\sigma\sqrt{\Delta t}}$. The tree recombines, reducing the number of nodes.

Backward induction

At each node discount the risk-neutral expected continuation value. For American exercise, compare continuation with immediate exercise.

Example

For $S_0=100, K=100, T=1, r=5\%, \sigma=20\%, q=0$ and $N=2$, compute u, d, p^* , terminal call payoffs and roll backward. Increase N and compare with Black-Scholes.

Convergence

As N increases, the discrete model should approach the continuous benchmark under matching assumptions. Convergence can be non-monotonic for some tree/payoff combinations.

Interview

Why use binomial? It makes replication and risk-neutral pricing transparent and naturally handles early exercise.

5. Risk-Neutral Pricing

Physical measure

Under the real-world measure P , a basic stock model has $dS = \mu S dt + \sigma S dW^P$.

Risk-neutral measure

Under Q , $dS = (r - q)S dt + \sigma S dW^Q$ in the basic constant-rate model.

Pricing equation

$V_0 = e^{-rT} E^Q[V_T]$. More generally, discounting must be consistent with the numeraire and rate model.

Why μ disappears

A dynamically hedged portfolio removes the diffusion risk. A locally riskless portfolio must earn the risk-free rate, so the physical drift is irrelevant to the arbitrage-free price in the complete model.

Martingale

Discounted tradable prices are martingales under Q in the standard setup. This is a pricing statement, not a claim about investor beliefs.

Interview

Is Q the actual probability distribution? No. It is a pricing measure constructed so that no-arbitrage valuation works.

6. Brownian Motion, GBM & Itô

Brownian motion

$W_0=0$; increments are independent and $W_t - W_s \sim N(0, t-s)$; paths are continuous but nowhere differentiable.

Stochastic rules

$(dW)^2 = dt$, $dW \cdot dt = 0$ and $(dt)^2 = 0$ in the Itô bookkeeping limit.

GBM

$dS = \mu S dt + \sigma S dW$. Its exact solution is $S_T = S_0 \exp[(\mu - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T} Z]$.

$$dS = \mu S dt + \sigma S dW$$

Risk-neutral GBM

$S_T = S_0 \exp[(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T} Z]$, $Z \sim N(0,1)$.

Itô lemma

For $dX = a dt + b dW$: $df = [f_t + a f_x + \frac{1}{2}b^2 f_{xx}]dt + b f_x dW$.

$$df = [f_t + a f_x + \frac{1}{2}b^2 f_{xx}]dt + b f_x dW$$

Applied to option

$dV = [V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS}]dt + \sigma S V_S dW$.

Why second derivative survives

The random increment is $O(\sqrt{dt})$, so its square is $O(dt)$. Thus the second-order Taylor term is first-order in dt .

Interview

What is the hardest conceptual difference from ordinary calculus? Brownian motion has non-zero quadratic variation, producing the second-derivative term.

7. Black-Scholes-Merton

Hedge derivation

Set $\Pi = V - \Delta S$. From Itô, choose $\Delta = V_S$ to cancel the dW term. Equate the locally riskless return to r .

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + (r - q)SV_S - rV = 0$$

PDE

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + (r - q)SV_S - rV = 0.$$

Call

$$C = S e^{-qT} N(d_1) - K e^{-rT} N(d_2).$$

$$C = S \exp(-qT) N(d_1) - K \exp(-rT) N(d_2)$$

d_1/d_2

$$d_1 = [\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T] / (\sigma\sqrt{T}); \quad d_2 = d_1 - \sigma\sqrt{T}.$$

$$d_1 = [\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T] / (\sigma\sqrt{T}); \quad d_2 = d_1 - \sigma\sqrt{T}$$

Put

$$P = K e^{-rT} N(-d_2) - S e^{-qT} N(-d_1).$$

Interpretation

$N(d_1)$ is call Delta adjusted for dividends. $N(d_2)$ has a risk-neutral exercise-probability interpretation. Neither is simply the real-world probability.

Assumptions

Constant volatility and rates; frictionless continuous trading; continuous dividend yield; no jumps; idealized liquidity and shorting assumptions.

Exceptions

American exercise, discrete dividends, stochastic volatility/rates, jumps and transaction costs require modifications or different models.

Interview

Why is Black-Scholes still important? It is the benchmark from which implied volatility, Greeks and many more advanced models are defined.

8. Greeks

Delta

$\Delta = \partial V / \partial S$. Call Delta = $e^{-qT} N(d_1)$; put Delta = $e^{-qT} [N(d_1) - 1]$. It is the instantaneous hedge ratio.

$\Delta = e^{-qT} N(d_1)$ for a call; $\Delta = e^{-qT} [N(d_1) - 1]$ for a put

Gamma

$\Gamma = \partial^2 V / \partial S^2 = e^{-qT} \phi(d_1) / (S\sigma\sqrt{T})$ for vanilla European call/put. It measures curvature and Delta instability.

$\Gamma = e^{-qT} \phi(d_1) / (S\sigma\sqrt{T})$

Vega

Vega = $\partial V / \partial \sigma = S e^{-qT} \phi(d_1) \sqrt{T}$. Long vanilla options are generally positive Vega.

Vega = $S e^{-qT} \phi(d_1) \sqrt{T}$

Theta

Theta is $\partial V / \partial t$ in one convention; systems may quote daily decay with a different sign. Always document convention.

Rho

Rho = $\partial V / \partial r$. Call and put signs differ.

Qualitative behavior

Long call: +Delta, +Gamma, +Vega, usually -Theta. Long put: -Delta, +Gamma, +Vega, usually -Theta.

Gamma concentration

Gamma tends to be highest around ATM and can become very large near expiry. This is why short-dated options can require frequent hedging.

P&L; approximation

$dV \approx \Delta dS + \frac{1}{2} \Gamma (dS)^2 + \Theta dt + \text{Vega } d\sigma + \text{Rho } dr + \dots$. It is local, not exact for large moves.

Interview

Why do calls and puts with the same inputs have the same Gamma? Gamma is the curvature of the common nonlinear option-value relationship; the parity difference is linear in S.

9. Numerical Greeks

Central difference

$\Delta \approx [V(S+h) - V(S-h)] / (2h)$. $\Gamma \approx [V(S+h) - 2V(S) + V(S-h)] / h^2$.

Step-size trade-off

Large h causes truncation error; tiny h causes cancellation and floating-point/noise error. Test sensitivity to h .

Monte Carlo

Finite-difference Greeks on noisy MC prices can be unstable. Common random numbers can reduce the variance of price differences.

Validation

Compare analytical and numerical Greeks across a grid, not one point. Report absolute and relative error.

Interview

Why is Gamma numerically harder than Delta? It uses a second difference divided by h^2 , amplifying both approximation and numerical noise.

10. Monte Carlo Pricing

Estimator

$$V_{\text{hat}} = e^{-rT} (1/N) \sum \text{payoff}_i$$

$$V_{\text{hat}} = e^{-rT} (1/N) \sum \text{payoff}_i$$

Simulation

For a European payoff under risk-neutral GBM: $S_T = S_0 \exp[(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z]$.

Standard error

$$SE \approx \text{sample_std}(\text{discounted payoff}) / \sqrt{N}$$

$$SE \approx \text{sample_std}(\text{discounted payoff}) / \sqrt{N}$$

Confidence interval

Approximate 95% CI: $V_{\text{hat}} \pm 1.96SE$ when the normal approximation is appropriate.

Convergence

Sampling error decreases as $O(1/\sqrt{N})$. Ten times lower error needs roughly 100 times more paths.

Antithetic variates

Simulate Z and $-Z$ and average paired payoffs. This can reduce variance when the payoff mapping has suitable dependence.

Control variates

Use a related variable with known expectation and adjust the estimator using the observed deviation.

Path-dependent options

Asian/barrier products require time grids and introduce discretization error as well as sampling error.

Exceptions

MC is inefficient for simple European vanilla options with closed forms, but attractive for path-dependent/high-dimensional products.

Interview

How do you validate MC? Fix a seed for reproducibility, test convergence with N , report SE/CI, compare to a closed-form benchmark and check discounting/simulation equations.

11. Implied Volatility

Definition

Solve $f(\sigma) = \text{ModelPrice}(\sigma) - \text{MarketPrice} = 0$.

Bisection

Requires a valid bracket; robust but relatively slow.

Newton

$\sigma_{\text{new}} = \sigma - f(\sigma) / f'(\sigma)$, with $f'(\sigma) = \text{Vega}$.

$$\sigma_{\text{new}} = \sigma - f(\sigma) / \text{Vega}$$

Example

If $\sigma = 20\%$, model price = 10.20, market = 10.00 and Vega = 35, then $\sigma_{\text{new}} \approx 0.20 - 0.20/35 \approx 19.43\%$.

Failure modes

No root because market price violates bounds; poor initial guess; very low Vega; numerical issues; inconsistent market data.

Practical solver

Check price bounds first. Use a bracketed method or Newton with robust fallback. Record status, iterations, residual and final tolerance.

Interview

Why can implied volatility be undefined? Because not every observed price is compatible with the model's arbitrage-free price range.

12. Volatility Smile, Skew & Surface

Definition

Implied volatility is a function $\sigma_{\text{impl}}(K, T)$, not a single market constant.

Smile/skew

A smile is curved variation with strike; skew is asymmetric variation. Equity markets often show higher downside implied volatility.

Reasons

Fat tails, skewness, stochastic volatility, jumps, supply/demand, risk premia and liquidity.

Surface

Organize IV across strike/moneyness and maturity. Interpolation must be chosen carefully to avoid unstable Greeks or arbitrage issues.

Extensions

Know at a high level why local volatility, Heston/stochastic volatility and SABR-type models are used.

Interview

Why does a constant-volatility model still matter? It is the benchmark language for option markets and gives a clean baseline for model comparison.

13. Delta Hedging

Basic hedge

For a long option with Delta 0.65, hedge with approximately -0.65 units of underlying per unit option exposure.

Continuous theory

Black-Scholes assumes continuous rebalancing; under its assumptions the hedge replicates the option.

Discrete reality

Real rebalancing leaves Gamma/Theta exposure and produces hedge error. Jumps and transaction costs further break ideal replication.

Simulation

At each hedge date: update S , revalue option, calculate Delta, rebalance underlying, accrue cash at the chosen funding rate, and settle payoff at maturity.

P&L;

Track both stock position and cash account. A correct self-financing hedge cannot be represented by stock-position changes alone.

Frequency experiment

Compare daily, weekly and monthly hedging. Study hedge-P&L; mean, standard deviation, tails and transaction costs.

Transaction costs

More frequent hedging reduces model hedge error but increases trading costs; therefore “more frequent” is not automatically optimal.

Interview

Why can a Delta-hedged option still lose money? Delta is first-order only; Gamma, Theta, volatility mismatch, jumps, rates, discrete hedging and costs remain.

14. Gamma, Hedge Error & Realized Volatility

Local P&L;

After Delta hedging, residual P&L; contains approximately $\frac{1}{2}\Gamma(dS)^2 + \Theta dt$ plus volatility/rate effects and higher-order terms.

Gamma scalping intuition

Long Gamma benefits from movement and rebalancing convexity but pays Theta. Profitability depends on realized movement relative to implied volatility and costs.

Realized vs implied

A Delta-hedged position is sensitive to realized variance relative to the variance embedded in the option, but exact P&L; depends on path, Gamma profile, hedging schedule, rates, dividends, jumps and costs.

Interview

Why does high Gamma require frequent hedging? Because Delta changes rapidly when S moves, so an old hedge becomes inaccurate quickly.

15. Calibration

Objective

$\theta^* = \operatorname{argmin} \sum w_i [P_{\text{model},i}(\theta) - P_{\text{market},i}]^2$.

Exact vs optimization

One parameter/one quote can be a root problem; many instruments and parameters normally require optimization.

Weights

Weights may reflect liquidity, bid-ask spread, price scale or business priorities.

Diagnostics

Report residuals, objective, parameters, convergence status, starting values and sensitivity.

Pitfalls

Market noise, inconsistent quotes, ill-conditioning, local minima, overfitting and poor weighting.

Project path

Start with option-by-option Black-Scholes IV; then construct an IV surface and only later add parametric/stochastic-volatility calibration.

Interview

Calibration fits parameters; validation tests whether the model and implementation remain reliable beyond the fitting exercise.

16. Model Validation

Financial invariants

Put-call parity, price bounds, monotonicity, limiting cases and known benchmarks.

Cross-model

Binomial European price should converge toward BS as steps increase. MC should agree with the benchmark within statistical uncertainty.

Regression tests

Keep trusted benchmark cases and verify every code change within documented tolerances.

Property tests

Test that call price generally increases with S and σ , decreases with K , and approaches intrinsic value as $T \rightarrow 0$ under standard assumptions.

Error taxonomy

Model error = bad assumptions; implementation error = wrong code; discretization error = numerical approximation; sampling error = finite MC paths; market-data error = bad/stale inputs.

Unit/integration

Unit-test payoffs, discount factors, $d1/d2$, CDF, Greeks and solvers. Integration-test the full pricing $\rightarrow$ IV $\rightarrow$ hedging workflow.

Interview

A plausible number is not validation. Independent identities and convergence tests are stronger evidence.

17. Numerical Stability & Performance

Floating point

Subtraction of nearly equal numbers can lose precision. This affects finite differences and extreme option values.

Small maturity

$T \rightarrow 0$ creates sharp sensitivities and division by $\sqrt{T}$. Handle expiry directly.

Zero volatility

$\sigma=0$ makes the standard BS formula singular; use the deterministic limiting payoff.

Deep ITM/OTM

Absolute error may be more informative than relative error when the price is tiny.

Vectorization

Use NumPy/vectorized operations for large MC workloads where practical; benchmark rather than assuming performance.

Reproducibility

Use fixed seeds in tests and separate deterministic validation from stochastic experiments.

Complexity

Basic BS is $O(1)$ per option; a basic N -step tree has $O(N^2)$ nodes; MC cost scales with paths and time steps.

Interview

Why can a smaller finite-difference h be worse? It reduces truncation error but can amplify floating-point cancellation and pricing noise.

18. Edge Cases & Exceptions

T=0

Return the payoff directly; do not evaluate $d1/d2$.

$\sigma=0$

Return the deterministic limit under the selected carry model.

Invalid market quote

If the price violates no-arbitrage bounds, IV may not exist. Return a diagnostic rather than forcing a solution.

American option

Do not use the vanilla European BS closed form without discussing early exercise. Binomial/PDE methods are common choices.

Discrete dividends

A continuous dividend yield may be inadequate when dividends are material and known discretely.

Negative rates

Models can be adapted, but implementation conventions and numerical behavior must be tested.

Theta convention

Document whether theta is $\partial V / \partial t$, per-year, per-day, and its sign convention.

Monte Carlo disagreement

First distinguish sampling error, discretization error, market-input differences and actual coding bugs.

Interview

What should an engine do when IV has no solution? Return a clear failure/status code with bounds, residuals and solver diagnostics.

19. Project Architecture & Implementation

Modules

payoffs; market data/input validation; black_scholes; binomial; monte_carlo; greeks; implied_vol; hedging; calibration; validation; diagnostics.

Input validation

$S > 0, K > 0, T \geq 0, \sigma \geq 0$; positive path/step counts; consistent units and rates.

Pricing interface

Use explicit contract/market/model objects rather than long positional argument lists. Return price plus diagnostics where useful.

Validation dashboard

Show benchmark prices, cross-model errors, convergence, Greek comparisons, IV solver iterations, hedge-P&L; distribution and runtime.

Recommended workflow

Inputs → validation → BS → Greeks → binomial → MC → IV → hedge simulation → calibration/extension → validation report.

Interview

Explain not only what you coded, but why the architecture separates pricing, numerical methods and validation.

20. High-Value Quant Interview Bank

Interview: Derive put-call parity.

Answer: Construct two portfolios with identical terminal cash flows and apply no-arbitrage.

Interview: Derive the one-step binomial hedge.

Answer: Solve two equations for Δ and the cash position B .

Interview: What is risk-neutral probability?

Answer: A pricing probability under Q ; it is not necessarily the real-world probability.

Interview: Why does μ disappear from Black-Scholes?

Answer: The Delta hedge removes the diffusion term; the locally riskless portfolio must earn r .

Interview: Derive the BS PDE.

Answer: Itô on V , form $V - \Delta S$, set $\Delta = V_S$, eliminate dW and equate drift to r .

Interview: Why is $(dW)^2 = dt$?

Answer: It is the quadratic-variation property used in Itô calculus.

Interview: What is $d1$?

Answer: A standardized model quantity in the closed-form solution; $N(d1)$ is linked to call Delta.

Interview: Is $N(d2)$ the real probability of exercise?

Answer: No. It has a risk-neutral probability interpretation, not a physical-probability interpretation.

Interview: Why is Gamma positive for long vanilla options?

Answer: Because the option value is convex in the underlying.

Interview: Why does Gamma increase near expiry ATM?

Answer: The option's Delta transitions rapidly around the strike as time becomes short.

Interview: Why is Vega usually positive?

Answer: More volatility increases the value of convex optionality.

Interview: Why is Monte Carlo slow?

Answer: Sampling error decreases only as $1/\sqrt{N}$.

Interview: How would you reduce MC variance?

Answer: Antithetic variates, control variates, importance sampling, quasi-random methods and better estimators.

Interview: Why use binomial over BS?

Answer: It is intuitive for replication and can handle early exercise via backward induction.

Interview: Why can Newton IV fail?

Answer: No root, poor starting value, low Vega, unstable update or inconsistent quote.

Interview: Calibration vs validation?

Answer: Calibration fits parameters; validation tests assumptions, implementation and robustness.

Interview: What is model risk?

Answer: Risk from incorrect assumptions, parameters, implementation or inappropriate use.

Interview: What is hedge error?

Answer: Residual P&L; from discrete rebalancing, nonlinear exposure, model mismatch, jumps, rates and costs.

Interview: How do you validate a pricing engine?

Answer: Use identities, bounds, limiting cases, independent benchmarks, convergence, property tests and regression tests.

Interview: What would you present in a project interview?

Answer: Architecture, derivations, assumptions, benchmark results, convergence, error diagnostics, hedge-P&L; analysis and limitations.

21. Final Study Checklist

- Probability: expectation, variance, covariance, conditional expectation, normal/lognormal distributions, LLN, CLT
- Calculus: partial derivatives, Taylor expansion, numerical differentiation
- No-arbitrage and cash-and-carry
- Forward pricing

- European/American option payoffs and moneyness
- Put-call parity and bounds
- One-step and multi-step binomial
- Risk-neutral valuation
- Brownian motion and GBM
- Itô's lemma
- Black-Scholes PDE and closed form
- Greeks and economic interpretation
- Finite-difference Greeks
- Monte Carlo, standard error and confidence intervals
- Variance reduction
- Implied volatility and root finding
- Volatility smile/skew
- Delta hedging and self-financing cash account
- Gamma/Theta/hedge error
- Calibration
- Model validation and testing
- Numerical stability and edge cases
- Project architecture and diagnostics

22. Formula Sheet

Topic	Formula
Forward	$F_0 = S_0 \exp((r-q)T)$
Call payoff	$\max(S_T - K, 0)$
Put payoff	$\max(K - S_T, 0)$
Parity	$C - P = S_0 e^{-qT} - K e^{-rT}$
CRR	$u = e^{\sigma\sqrt{\Delta t}}, d = e^{-\sigma\sqrt{\Delta t}}$
Risk-neutral p	$p^* = [e^{(r-q)\Delta t} - d] / (u - d)$
Risk-neutral value	$V_0 = e^{-rT} E^Q[V_T]$
GBM	$dS = \mu S dt + \sigma S dW$
BS PDE	$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + (r-q)SV_S - rV = 0$
Call BS	$C = S e^{-qT} N(d_1) - K e^{-rT} N(d_2)$
d1	$[\ln(S/K) + (r-q + \frac{1}{2}\sigma^2)T] / (\sigma\sqrt{T})$
d2	$d_1 - \sigma\sqrt{T}$
Delta	$\partial V / \partial S$
Gamma	$\partial^2 V / \partial S^2$
Vega	$\partial V / \partial \sigma$
Theta	$\partial V / \partial t$; document convention
Rho	$\partial V / \partial r$
MC	$V_{\text{hat}} = e^{-rT} (1/N) \sum \text{payoff}_i$
MC SE	$\text{sample_std}(\text{discounted payoff}) / \sqrt{N}$
Newton IV	$\sigma_{\text{new}} = \sigma - f(\sigma) / \text{Vega}$
Hedge	Underlying $\approx$ -Delta $\times$ option exposure

Topic	Formula
P&L	$dV \approx \Delta dS + \frac{1}{2} \Gamma (dS)^2 + \Theta dt + \text{Vega } d\sigma + \text{Rho } dr + \dots$